

Task 20: Implement branching strategies in Git by creating feature branches. Perform merging and rebasing operations, and intentionally create merge conflicts to understand and resolve them effectively.

AIM: To implement branching strategies in Git by creating feature branches, performing merging and rebasing operations, and intentionally creating merge conflicts to understand and resolve them effectively.

ALGORITHM:

1. Initialize a new Git repository
2. Create a main file and commit it
3. Create a feature branch
4. Modify the same file in both branches
5. Commit changes in both branches
6. Merge the feature branch into main (create conflict)
7. Resolve the merge conflict manually
8. Commit the resolved changes
9. Perform rebase operation
10. Handle conflicts during rebase
11. Complete the rebase and verify final output

CODE:

```
git init
```

```
echo Hello from Main Branch > file.txt
```

```
git add file.txt
```

```
git commit -m "Initial commit"
```

```
git checkout -b feature
```

```
echo Hello from Feature Branch > file.txt
```

```
git add file.txt
```

```
git commit -m "Feature branch change"
```

```
git checkout master
```

```
echo Hello from Master Branch Version 2 > file.txt
```

```
git add file.txt
```

```
git commit -m "Master branch update"
```

```
git checkout feature
```

```
echo Hello from Feature Branch Version 2 > file.txt
```

```
git add file.txt
```

```
git commit -m "Feature branch update"
```

```
git checkout master
```

```
git merge feature
```

Resolution:

```
git add file.txt
```

```
git commit -m "Resolved merge conflict"
```

```
git checkout feature
```

```
git rebase master
```

Output :

1. Merge Conflict

```
C:\Users\HARITHA\git-task>git merge feature
Auto-merging file.txt
CONFLICT (content): Merge conflict in file.txt
Automatic merge failed; fix conflicts and then commit the result.
```

2. Git Log Output

```
C:\Users\HARITHA\git-task>git log --oneline --graph
* 80d7bbb (HEAD -> feature, master) Resolved merge conflict
| \
| * 1624f72 Feature branch update
| * 5d29f7d Main branch update
| * 569f7a0 Feature branch change
| | f31c11a Master branch update
| /
* e8de353 Initial commit
```

RESULT: Thus, Git branching strategies were successfully implemented by creating feature branches and performing merge and rebase operations. A merge conflict was created and resolved manually.